

## Arithmetic sequences



1 By choosing any term after the first and subtracting the preceding term from it.

2

a No

b No

3

a 20, 24,

b  $-10, -14$

c 8, 9.5

d  $-\frac{20}{3}, -\frac{19}{3}$

4

a  $d = -50$

b  $d = \frac{3}{7}$

5

a i 11, 14, 17, 20    ii 3

b i 11, 21, 31, 41    ii 10

c i  $-7, -10, -13, -16$     ii  $-3$

6

a  $d = 0.9$

b  $n = 11$

7  $-6, 0, 6, 12, 18$

8

a  $n = 37$

b  $n = 27$

c  $n = 117$

d  $n = 57$

9



a

i  $d = 7$

ii  $T_n = 7n - 2$

iii  $T_{10} = 68$

b

i  $d = -0.8$

ii  $T_n = -0.8n + 17.8$

iii  $T_{10} = 9.8$

c

i  $d = -7$

ii  $T_n = -7n + 17$

iii  $T_{10} = -53$

d

i  $d = \frac{3}{4}$

ii  $T_n = \frac{3n + 17}{4}$

iii  $T_{10} = \frac{47}{4}$

## Recursive rules

10  $T_n = T_{n-1} - 6, T_1 = 3$

11  $T_n = T_{n-1} + 25, T_1 = 30$

12

a  $d = 6$

b  $T_n = T_{n-1} + 6, T_1 = 2$

13

a  $d = 5$

b  $T_1 = 4$

c  $T_n = T_{n-1} + 5, T_1 = 4$

## Tables and graphs

14

a

|       |   |   |   |    |
|-------|---|---|---|----|
| $n$   | 1 | 2 | 3 | 4  |
| $T_n$ | 1 | 5 | 9 | 13 |

b  $d = 4$



15

a

|       |    |   |    |
|-------|----|---|----|
| $n$   | 1  | 3 | 5  |
| $T_n$ | -3 | 7 | 17 |

b  $d = 5$

c  $T_n = -8 + 5n$

d  $T_{18} = 82$

16

a  $d = 5$

b  $T_n = -9 + 5n$

c  $m = 5$

17

a

i  $d = 6$

ii  $T_n = 3 + 6n$

iii  $T_{15} = 93$

b

i  $d = -9$

ii  $T_n = 14 - 9n$

iii  $T_{10} = -76$

c

i  $d = \frac{9}{2}$

ii  $T_n = -\frac{1}{2} + \frac{9n}{2}$

iii  $T_{11} = 49$

d

i  $d = -8$

ii  $T_n = -2 - 8n$

iii  $T_{19} = -154$

e

i  $d = -6$

ii  $T_n = -4 - 6n$

iii  $T_{18} = -112$

18

|       |    |     |     |     |     |
|-------|----|-----|-----|-----|-----|
| $n$   | 1  | 2   | 3   | 4   | 5   |
| $T_n$ | -6 | -11 | -16 | -21 | -26 |

19

a  $d = 3$

b  $T_n = -1 + 3n$

c  $T_k = 1 - 3k$



## Applications

20  $n = 32$

21

a \$43 920

b  $\frac{m}{2} (82\,080 + 1920m)$  dollars

c  $y = 9$

22 \$2670

23

a

| Day | left to pave ( $\text{m}^2$ ) |
|-----|-------------------------------|
| 1   | 800                           |
| 2   | 750                           |
| 3   | 700                           |
| 4   | 650                           |
| 5   | 600                           |

b Linear decay

24

a 5800 phones

b 165 600 phones

25 1510 in

26

a  $d = 35$

b  $a = 151$

c  $n = 22$

27

a  $3 + n$

b  $\frac{n}{2} (7 + n)$  pipes

28

a  $6k + 2$  yd

b  $n(3n + 5)$  yd

c  $n = 10$

29



30

a 4.5 km

b 112.25 km

31

a 430 lb

b 5390 lb

c 26

32

a \$300

b \$2200

c \$1900

d  $V_{n+1} = V_n + 300, V_0 = 1900$

33

a \$20

b \$1200

c  $V_{n+1} = V_n - 20, V_0 = 1200$

34

a 50 m

b 200 m

c 500 m

35

a \$1972

b \$1860

c  $T_n = T_{n-1} - 4, T_0 = 2000$

36

a \$21

b  $T_n = T_{n-1} + 1.95, T_0 = 1.5$

37

a \$24.09

b  $V_{n+1} = V_n - 2.05, V_0 = 42.54$

38

a \$365

b  $u_n = u_{n-1} + 45, u_0 = 230$

c  $v_n = v_{n-1} + 45, v_0 = 460$

39

a

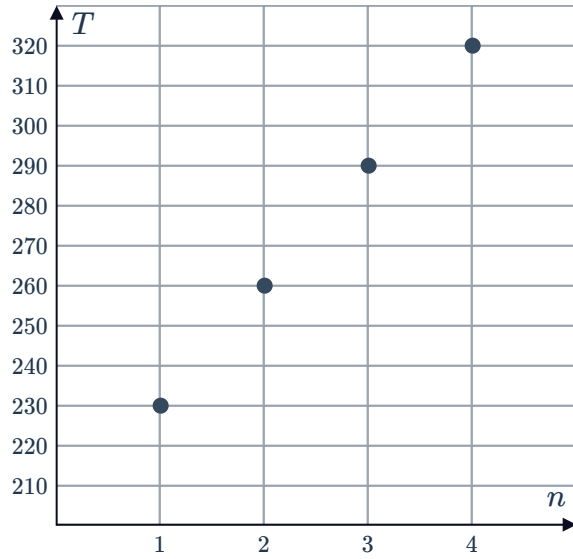
|     |     |     |     |     |     |
|-----|-----|-----|-----|-----|-----|
| $n$ | 1   | 2   | 3   | 4   | 18  |
| $T$ | 230 | 260 | 290 | 320 | 740 |



b

30

c



d A straight line